

# R Functions Lab (Class 06)

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**Q1. Write your first R function: add()**

```
## Q1. Write your first R function: add()
```

```
### Q1a
```

```
add <- function(x, y) {  
  x + y  
}
```

```
add(4, 7)
```

```
[1] 11
```

Q1b

```
add <- function(x, y = NULL) {  
  if (is.null(y)) {  
    sum(x)  
  } else {  
    x + y  
  }  
}
```

```
add(4, 7)
```

```
[1] 11
```

```
add(c(4, 7, 10))
```

```
[1] 21
```

Q1c

```
add <- function(...) {  
  sum(...)  
}
```

```
add(4, 7)
```

```
[1] 11
```

```
add(c(4, 7, 10))
```

```
[1] 21
```

```
add(1, 2, 3, -4)
```

```
[1] 2
```

**Q2. Write a generate\_dna() function**

```
generate_dna <- function(n) {  
  nts <- c("A", "T", "G", "C")  
  sample(nts, size = n, replace = TRUE)  
}
```

```
generate_dna(6)
```

```
[1] "T" "C" "C" "T" "T" "T"
```

Q2b

```
generate_dna <- function(n, as_string = TRUE) {
  nts <- c("A", "T", "G", "C")
  seq_vec <- sample(nts, size = n, replace = TRUE)

  if (as_string) {
    paste(seq_vec, collapse = "")
  } else {
    seq_vec
  }
}

generate_dna(6, as_string = FALSE)
```

```
[1] "T" "A" "T" "C" "C" "A"
```

```
generate_dna(11)
```

```
[1] "ACACGTTTGAA"
```

Q2c

```
generate_dna <- function(n) {
  nts <- c("A", "T", "G", "C")
  seq_vec <- sample(nts, size = n, replace = TRUE)
  seq_string <- paste(seq_vec, collapse = "")

  cat(paste0(">len", n, "\n", seq_string, "\n"))
}

generate_dna(11)
```

```
>len11
TTGTTGGTTTT
```

**Q3. Write a generate\_protein() function**

```
generate_protein <- function(n) {
  aa <- c("A", "R", "N", "D", "C", "E", "Q", "G", "H", "I",
         "L", "K", "M", "F", "P", "S", "T", "W", "Y", "V")

  seq_vec <- sample(aa, size = n, replace = TRUE)
  paste(seq_vec, collapse = "")
}

generate_protein(6)
```

```
[1] "FMFWKD"
```

#### Q4. Generate random protein sequences of length 6 to 13

```
for (i in 6:13) {
  cat(paste0(">id.", i, "\n"))
  cat(generate_protein(i), "\n")
}
```

```
>id.6
PAMYEN
>id.7
AVFQPAQ
>id.8
IVSEGDDH
>id.9
KCKAVFWQK
>id.10
YITGWHQRMG
>id.11
QFPTWMRPPNA
>id.12
VHCKCIVDIDQM
>id.13
YAHMCVETWILY
```

#### Q5. BLASTp search against nr — are your peptides “unique in nature”?

Sequence generated utilized for BLAST results:

>id.6  
 TDAWPL  
 >id.7  
 YGLNYKY  
 >id.8  
 ARNKIRME  
 >id.9  
 IHRNVMDTW  
 >id.10  
 GCLIHQYVT  
 >id.11  
 MLACGVSTCHI  
 >id.12  
 RCCESASQEEKI  
 >id.13  
 NEGICNLWGQSTH

### Results Table

| Length (aa) | Best hit % identity | Best hit % coverage | Unique? (Y/N) |
|-------------|---------------------|---------------------|---------------|
| 6           | 100%                | 100%                | N             |
| 7           | 100%                | 100%                | N             |
| 8           | 100%                | 100%                | N             |
| 9           | 87.5%               | 89%                 | Y             |
| 10          | 90.0%               | 90%                 | Y             |
| 11          | 90.0%               | 91%                 | Y             |
| 12          | 66.7%               | 100%                | Y             |
| 13          | 88.9%               | 69%                 | Y             |

Q5a: Random peptides will appear “unique in nature” when approaching the approximate 9 amino acid length. Shorter sequences have displayed a trend in perfect matches, while longer sequences usually did not.

Q5b: Peptides very short in length are frequently found because the total sequence space utilized is small. The number of possible sequences is  $20^L$ , so as length begins to increase, the number of possible sequences grows exponentially with it. This begins to exceed the number of known proteins very fast, making exact matches unlikely.

### Q6. Connecting your findings to immunology (MHC class II and T-cell activation)

The minimum peptide length required for MHC class II presentation is most likely to be around 9–12 amino acids. Shorter peptides are not a great choice because they are too common in

length and may appear in both self and non-self proteins. This would make it hard for the immune system to distinguish between them. Longer peptides occupy a much larger sequence space, making them more unique and better suited for identifying foreign antigens. This improves the specificity of immune recognition and helps ensure appropriate T-cell activation.